

Experiment # 06

Setting up Virtual Local Area (VLAN) Network

Objective:

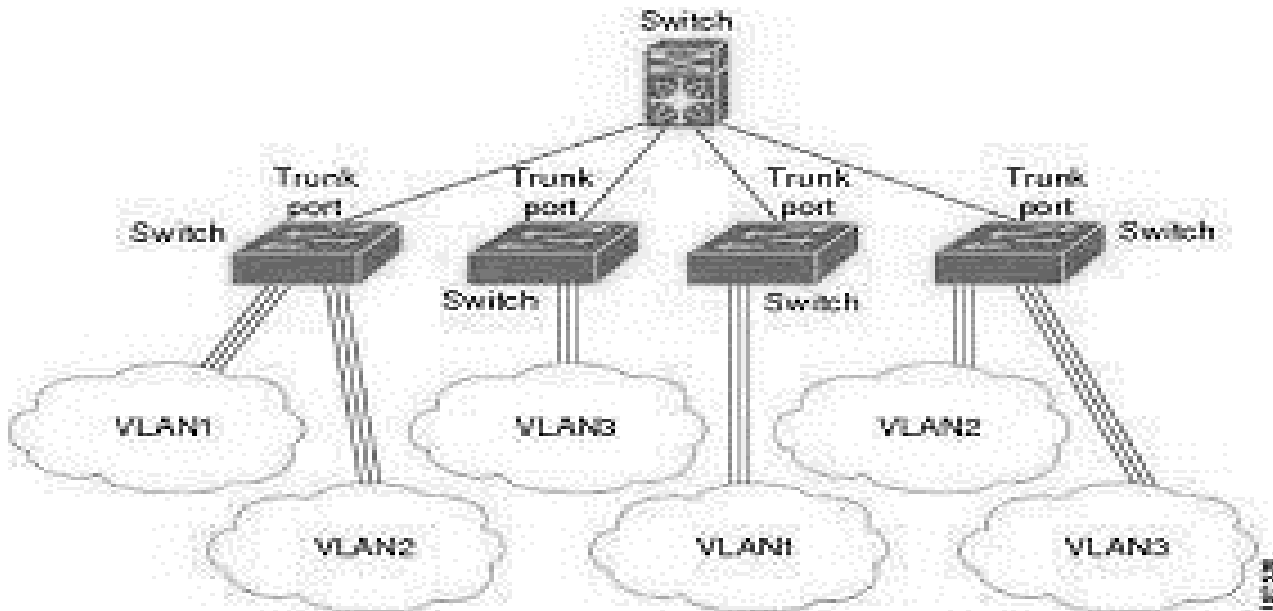
The objective of this lab is how to make VLAN using aggregate switch.

Software Tool:

- Cisco Packet Tracer

Theory:

In computer networking, a single network maybe partitioned to create multiple distinct broadcast domains, which are mutually isolated so that packets can only pass between them via one or more routers; such a domain is referred to as a virtual local area network, virtual LAN or VLAN. This is usually achieved on switch or router devices. Simpler devices only support partitioning on a port level (if at all), so sharing VLANs across devices requires running dedicated cabling for each VLAN. More sophisticated devices can mark packets through tagging, so that a single interconnect (trunk) may be used to transport data for multiple VLANs. Grouping hosts with a common set of requirements regardless of their physical location by VLAN can greatly simplify network design. A VLAN has the same attributes as a physical local area network (LAN), but it allows for end stations to be grouped together more easily even if they are not on the same network switch. VLAN membership can be configured through software instead of physically relocating devices or connections. Most enterprise-level networks today use the concept of virtual LANs. Without VLANs, a switch considers all interfaces on the switch to be in the same broadcast domain.



Packets destined for stations that do not belong to the VLAN must be forwarded through a router.

Advantages of VLAN

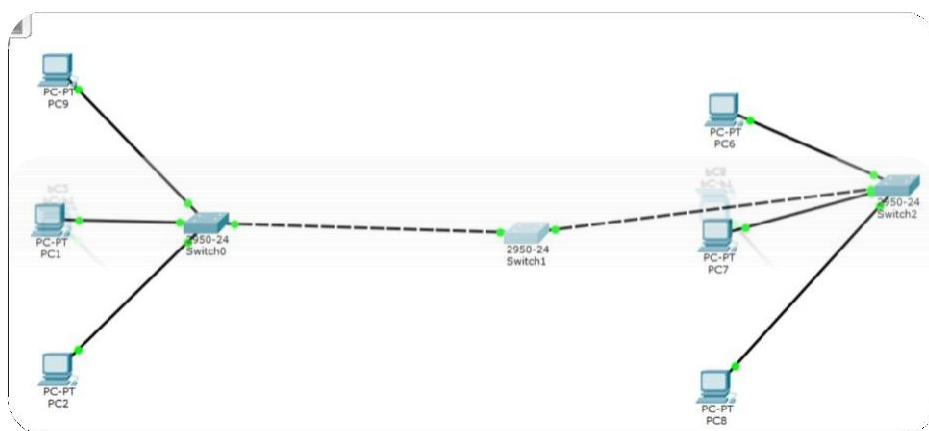
VLAN provides following advantages:-

- Solve broadcast problem
- Reduce the size of broadcast domains
- Allow us to add additional layer of security
- Make device management easier

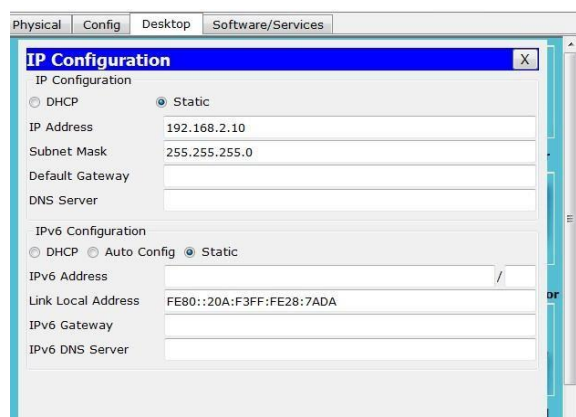
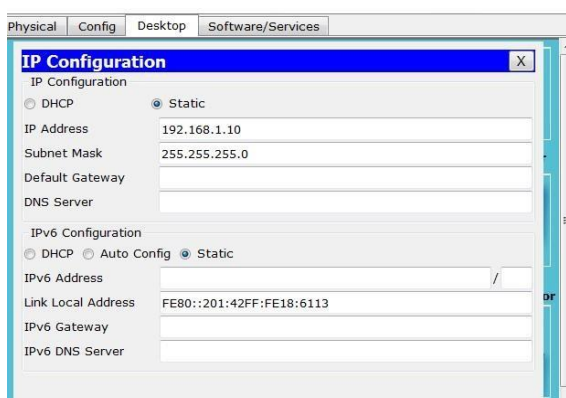
Allow us to implement the logical grouping of devices by function instead of location

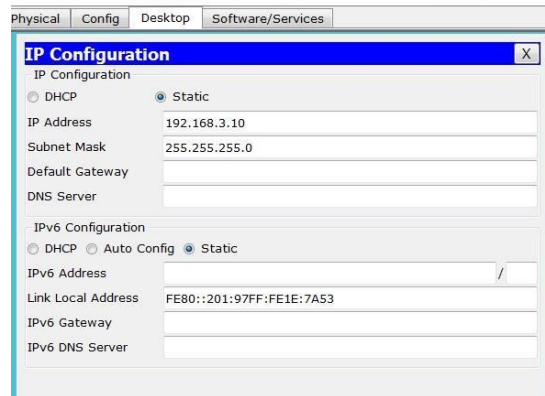
Procedure:

Connect three pc of same network but different IP addresses with a switch. And then connect this switch to another switch. After this take another switch and connect it with three pc of same IP but different networks.

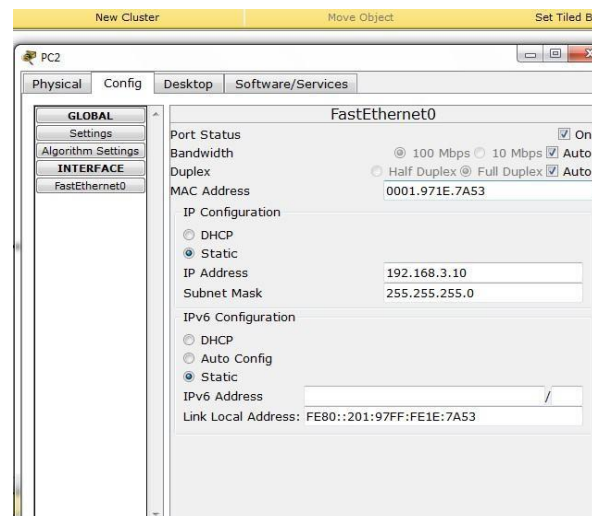
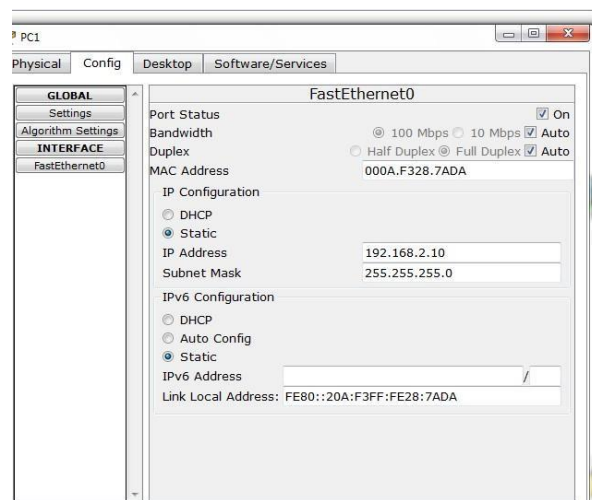
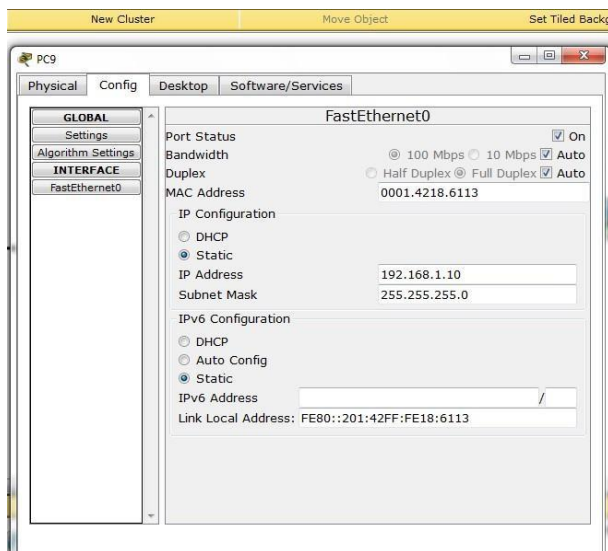


Now give IP's to pc0 pc1

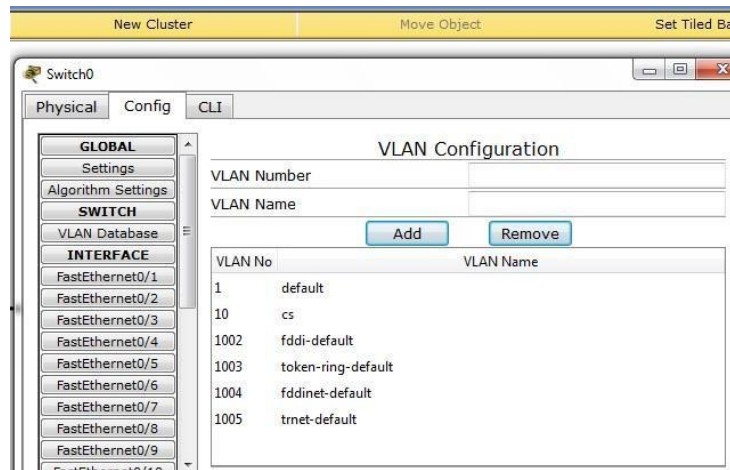




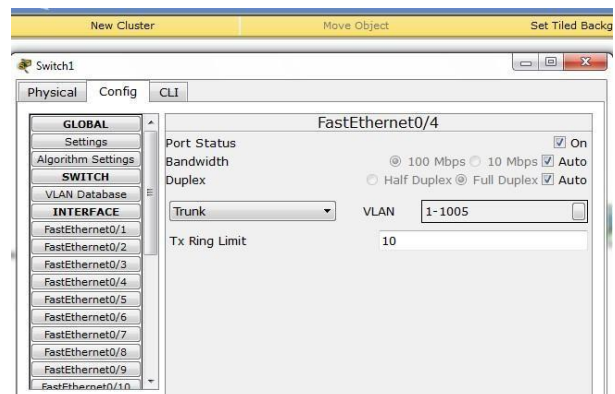
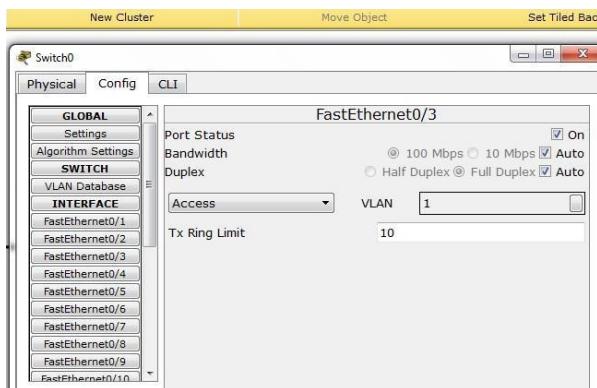
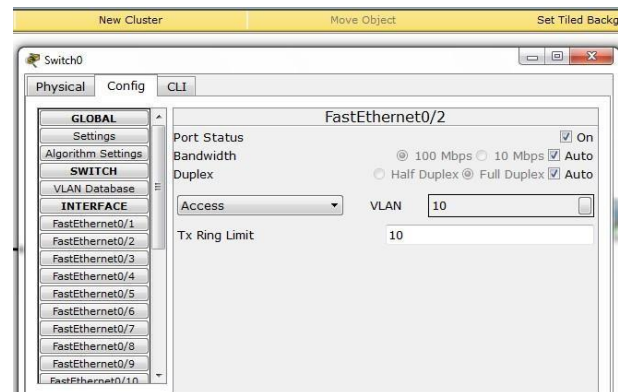
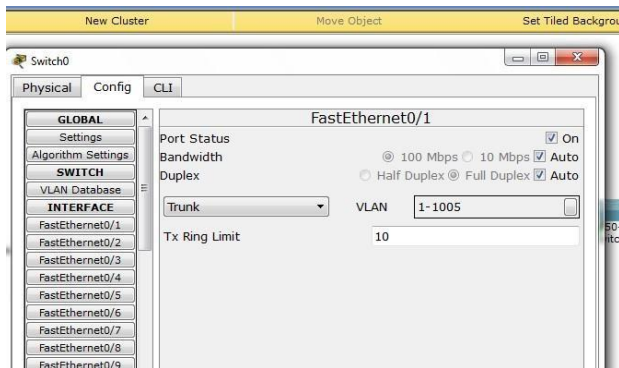
- Now set bandwidth on auto for pc0, pc1, pc2:



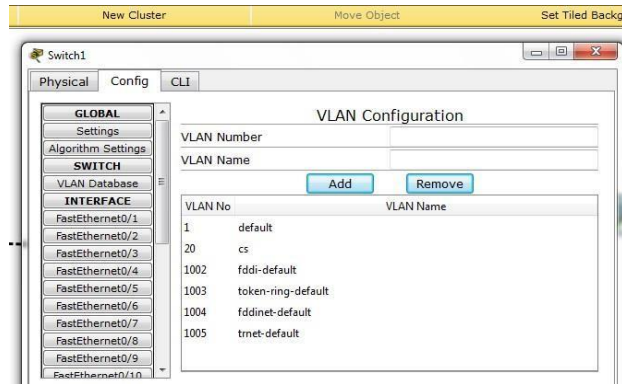
- First add the VLAN database in switch:



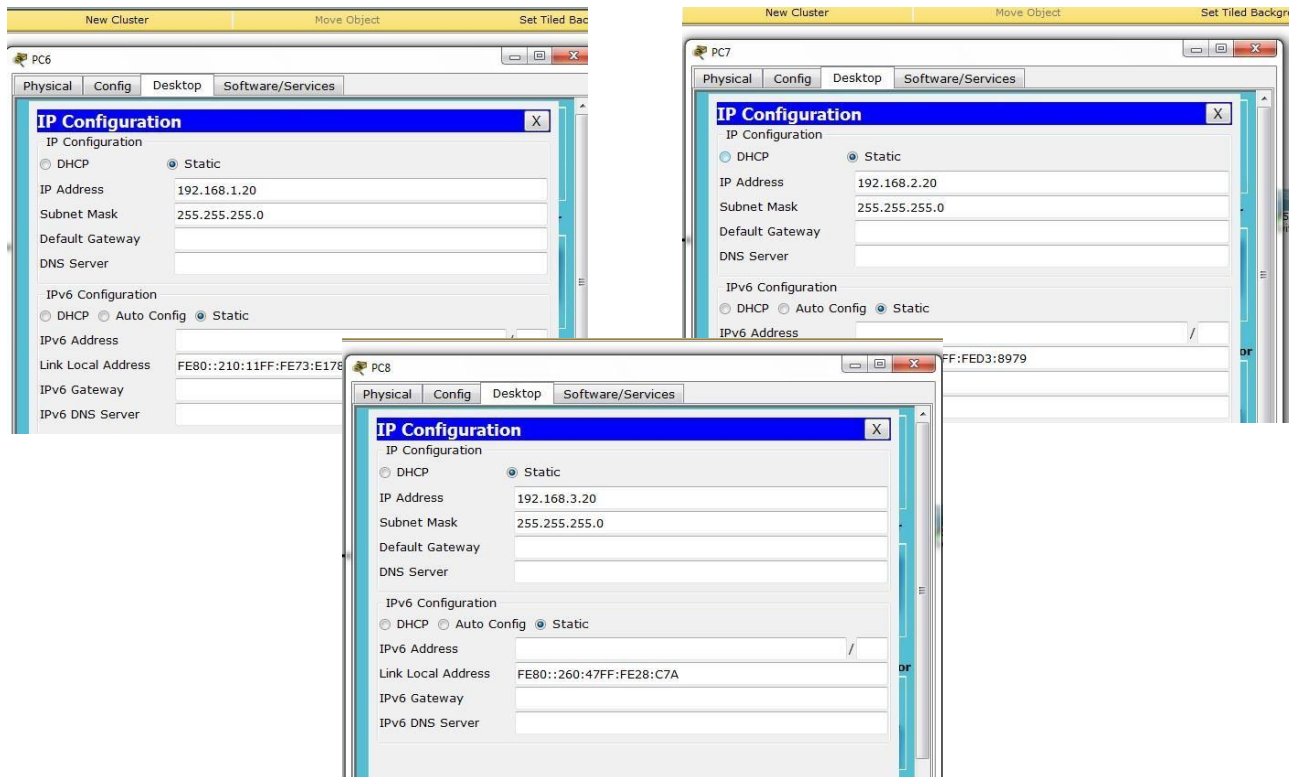
- And then set fast Ethernet 0/1 setting on trunk. First check which wire carries data in switch then set it as a trunk. And the other wires on access:



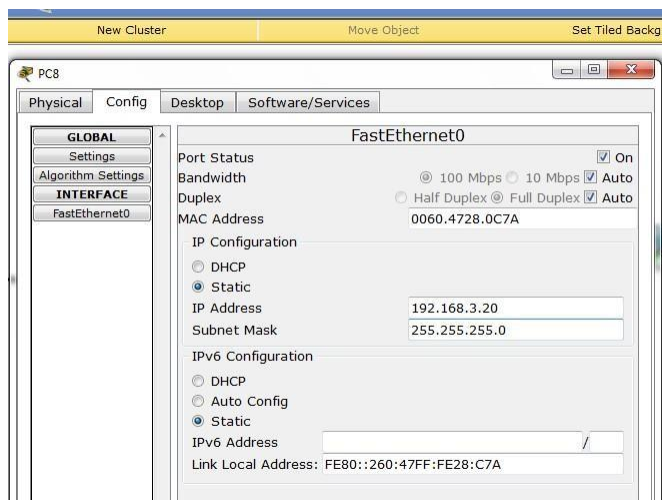
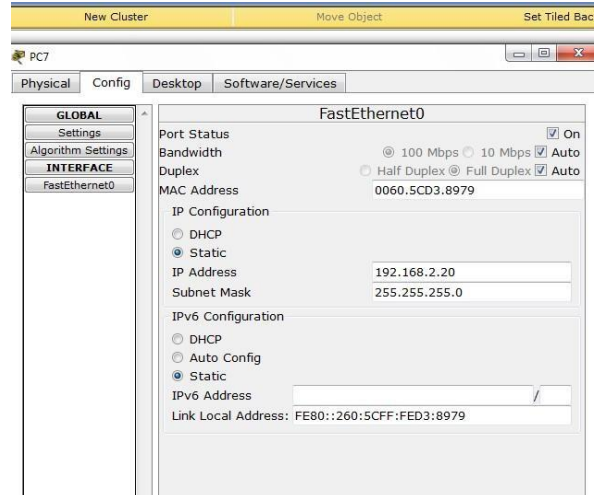
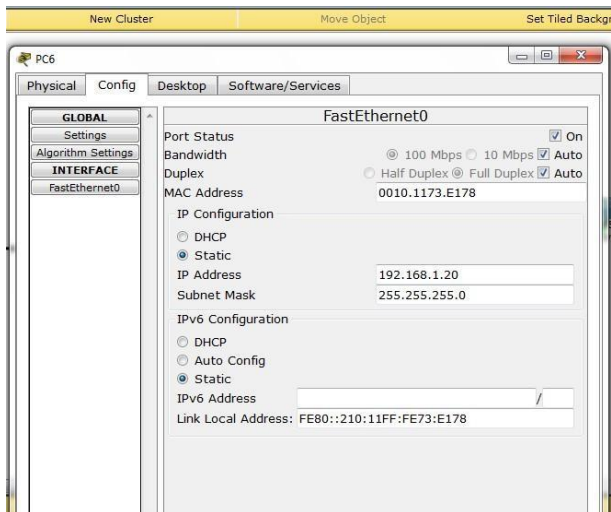
- Now add the VLAN database for switch 2. And set fast Ethernet 0/4 on trunk:



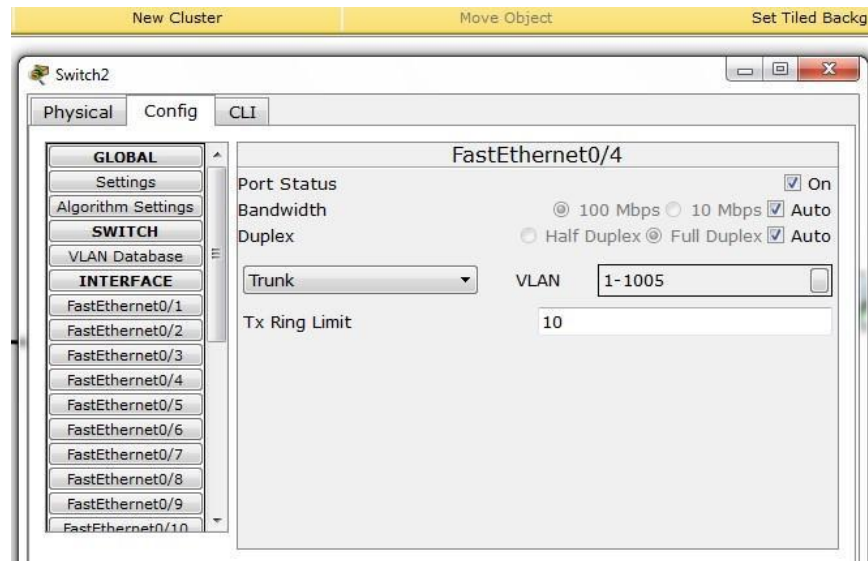
- Now give IP's to pc4, pc5, pc6:



- Now set the bandwidth for pc4, pc5, pc6 on auto:

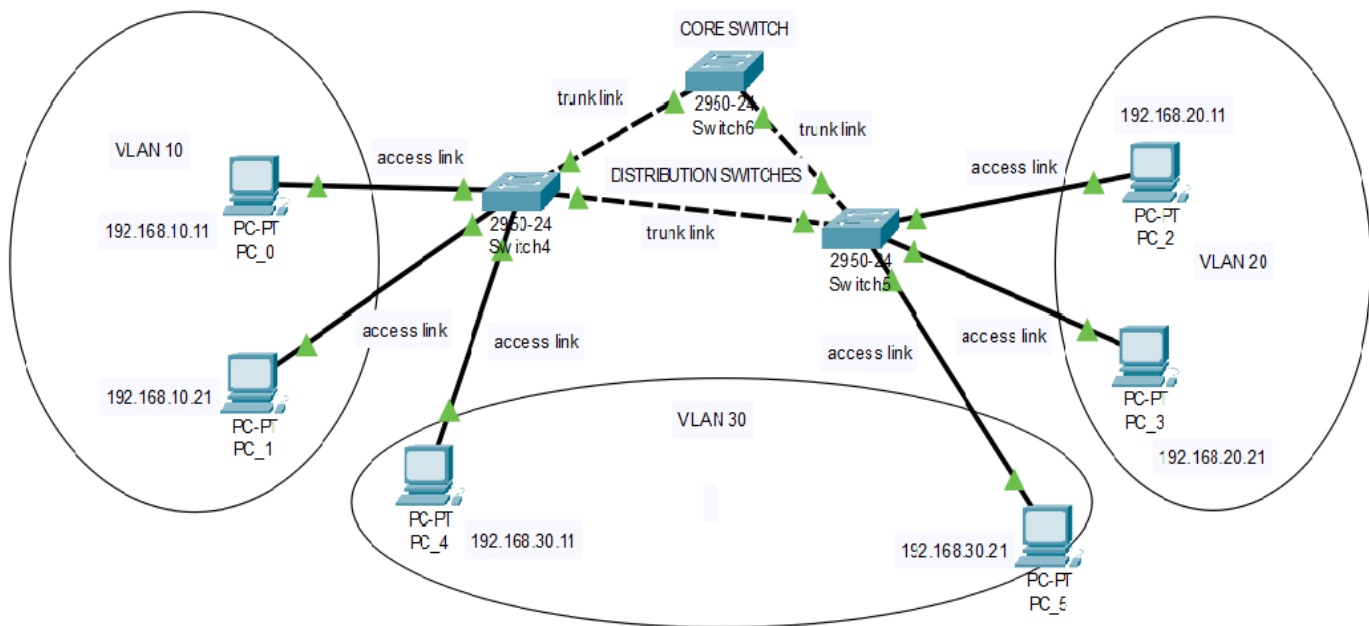


- Now add the VLAN database to switch 3 and set fast Ethernet 0/4 on trunk because it is connected to switch 2:

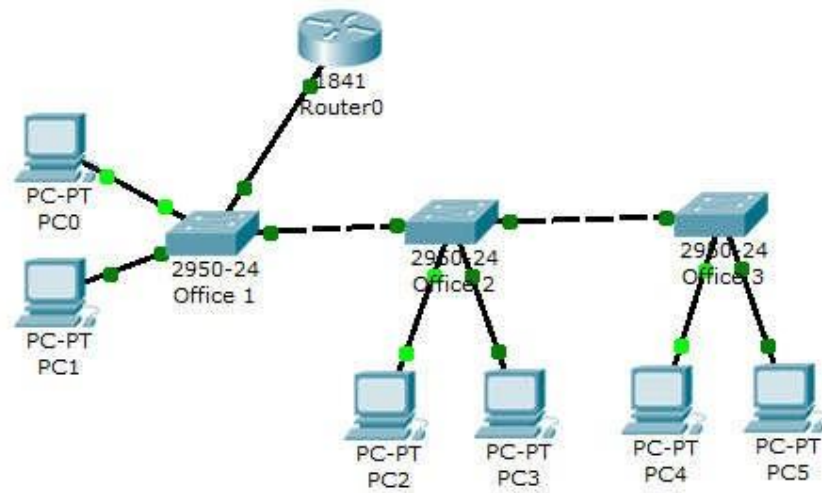


LAB TASK

1. **Simulate** following topology and implement intra-VLAN.



2. Simulate the following inter-VLAN:



Conclusion
